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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: Case Study Analysis

Case Study Title: Flood Early Warning System

(SDG 13:Climate Action)

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1. Introduction

Background of the Problem

Flooding is one of the most common natural disasters affecting many districts during heavy rainfall. Existing warning systems mainly depend on television, radio announcements, or manual communication, which often fail to reach residents quickly. Delayed warnings increase the risk of property damage, injuries, and loss of life.

The proposed **Flood Early Warning System** uses weather sensors, real-time data processing, machine learning, cloud computing, and mobile notifications to detect potential floods early. The system provides timely alerts, evacuation guidance, and emergency coordination between citizens, disaster management authorities, and rescue teams.

Importance of Solving the Problem

- Provides early flood detection
- Reduces loss of life and property
- Sends instant alerts to citizens
- Supports emergency rescue operations
- Improves disaster preparedness
- Enables quick evacuation planning
- Enhances public safety

Related SDG Goal

SDG 13 – Climate Action

Targets:

- Strengthen resilience against climate-related disasters
- Improve disaster preparedness
- Reduce disaster risks through technology
- Promote sustainable climate adaptation

Objectives of the Case Study

- Study the existing flood warning system.
- Identify current challenges.
- Design an intelligent flood monitoring software.
- Provide early warning notifications.
- Improve emergency response.
- Support SDG 13.

2. Problem Understanding

2.1 Problem Description

Currently, flood warnings are issued through television, radio, or local announcements. These methods cannot notify everyone immediately, especially people travelling or living in remote areas. Authorities also lack a centralized system for monitoring flood conditions in real time.

A software-based Flood Early Warning System can collect weather sensor data, predict flood risks using machine learning, generate alerts automatically, and provide evacuation guidance to citizens.

2.2 Objectives

- Monitor weather continuously
- Predict flood risks
- Generate early alerts
- Reduce disaster impact
- Support rescue teams
- Improve public safety
- Provide evacuation guidance
- Maintain historical flood records

2.3 Scope

In Scope

- Weather sensor monitoring
- Rainfall monitoring
- Water level monitoring
- Flood prediction
- Alert notification
- User login
- Evacuation guidance
- Disaster management dashboard
- Emergency coordination
- Report generation

Out of Scope

- Construction of dams
- Weather sensor manufacturing
- Physical rescue operations
- Government compensation process
- Insurance claim management

3. Stakeholder Analysis

Stakeholder	Role	Responsibilities
Administrator	System Admin	Manage users and system settings
Disaster Management Authority	Decision maker	Monitor flood status, issue alerts, coordinate rescue
Weather Sensor System	Data Provider	Collect rainfall, river level, humidity, and temperature data
Citizens	End Users	Receive alerts, check flood status, follow evacuation routes

Rescue Team	Emergency Response	Conduct rescue operations and provide assistance
Government	Regulatory Authority	Disaster policy implementation and monitoring

4. Existing System Analysis

4.1 Existing Process

Flood monitoring mainly depends on weather forecasts and manual observations. Information is communicated through television, radio, newspapers, and public announcements. These methods often cause delays, making evacuation difficult.

4.2 Strengths

- Simple communication methods
- Low technology requirements
- Government managed
- Familiar to the public

4.3 Weaknesses

- Delayed alerts
- Manual monitoring
- Limited public reach
- Lack of real-time monitoring
- No automatic prediction
- Poor emergency coordination
- Increased flood damage

4.4 Gap Analysis

Current system limitations include:

- No real-time weather monitoring
- No AI-based flood prediction
- No automatic notifications
- No mobile application
- No evacuation guidance
- No centralized dashboard
- No live sensor integration

5. Proposed Software Solution

5.1 Proposed System

The proposed Flood Early Warning System integrates weather sensors, cloud storage, machine learning models, and mobile applications. Weather sensors continuously monitor rainfall, water level, humidity, and temperature. The collected data is processed to predict flood risks. When flood conditions are detected, the system automatically sends alerts to citizens and disaster management authorities. Users can also view flood status and evacuation routes through the mobile application.

5.2 Functional Modules

- User Login
- Weather Monitoring
- Flood Prediction
- Alert Notification
- Flood Status Dashboard
- Evacuation Route Guidance
- Emergency Help
- Disaster Management Dashboard
- Report Generation
- System Administration

5.3 Functional Requirements

FR1: The system shall allow user login.

FR2: The system shall collect weather sensor data.

FR3: The system shall monitor rainfall and water levels.

FR4: The system shall predict flood risk.

FR5: The system shall send flood alerts.

FR6: The system shall display flood status.

FR7: The system shall provide evacuation routes.

FR8: The system shall generate reports.

FR9: The system shall notify disaster management authorities.

FR10: The system shall maintain historical weather data.

5.4 Non-Functional Requirements

- High Performance
- Reliability
- Scalability
- Security
- Availability
- Maintainability
- Portability
- Usability

5.5 Software Engineering Principles Applied

Modularity: Separate modules for monitoring, prediction, alerts, and reporting.

Scalability: Can support thousands of sensors across multiple districts.

Reliability: Continuous monitoring with secure cloud storage.

Security: User authentication and encrypted communication.

Maintainability: Easy to update individual modules.

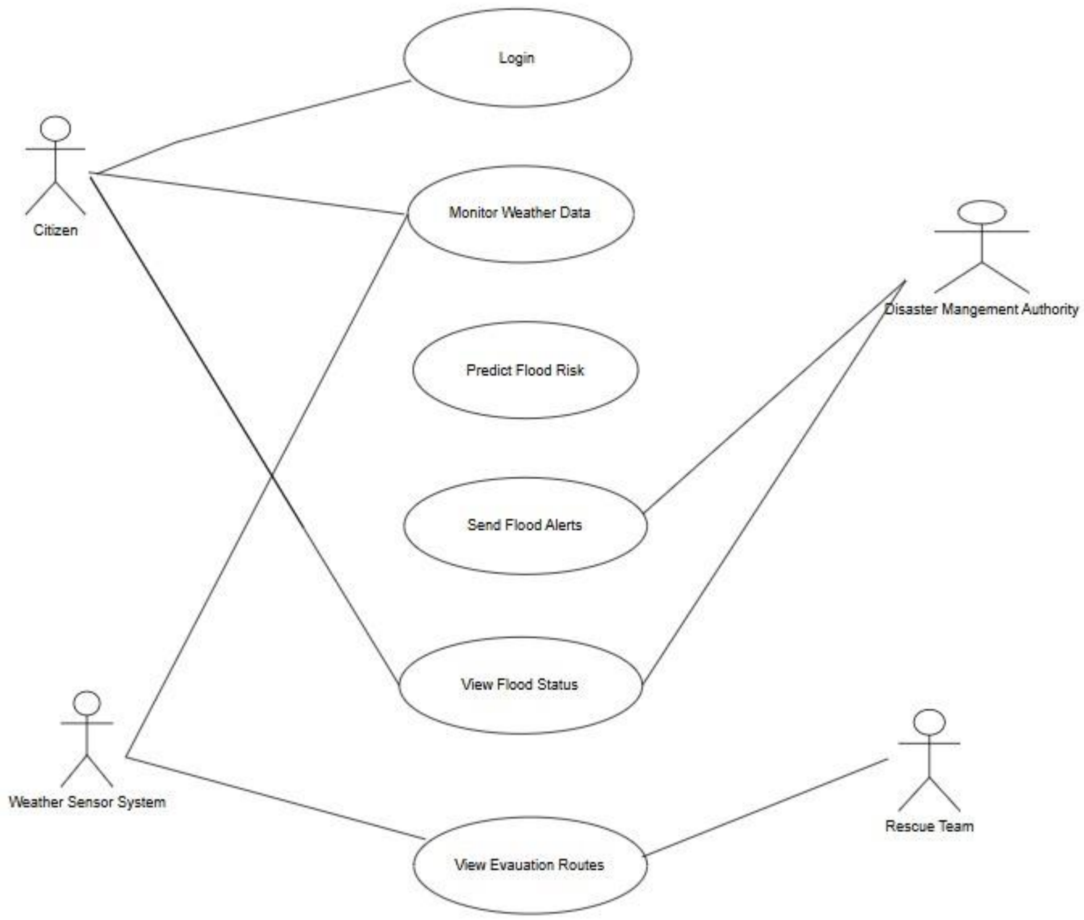
Reusability: Components can be reused in other disaster management applications.

Usability: User-friendly mobile application and dashboard.

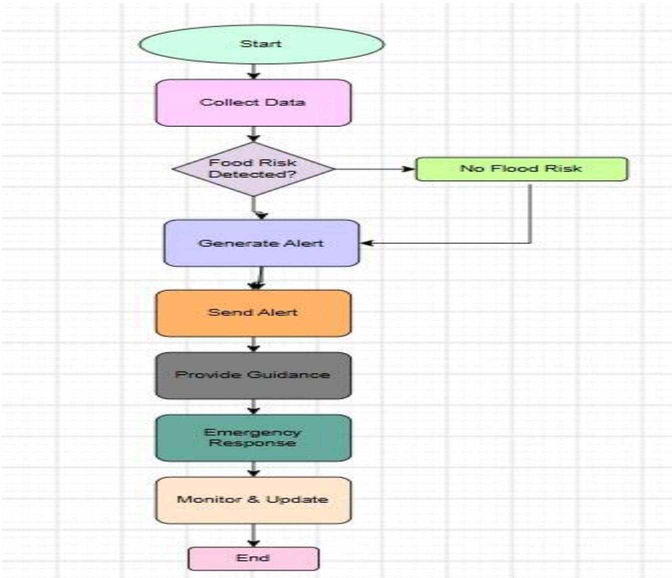
Flexibility: Supports future AI models and IoT devices.

6. System Design

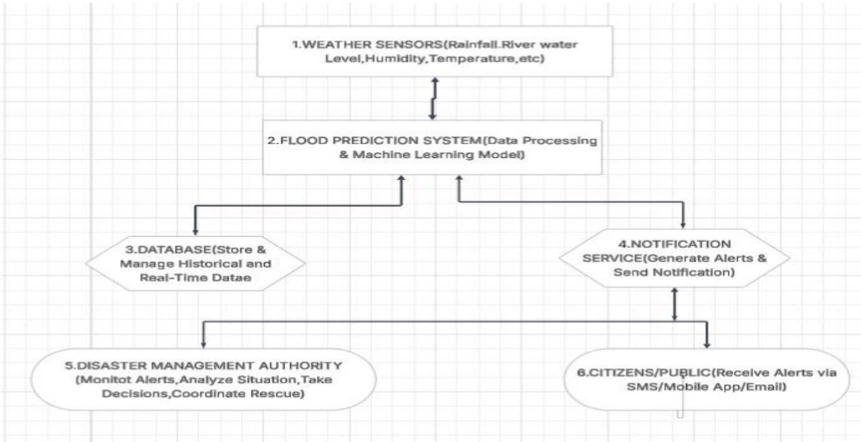
Use Case Diagram



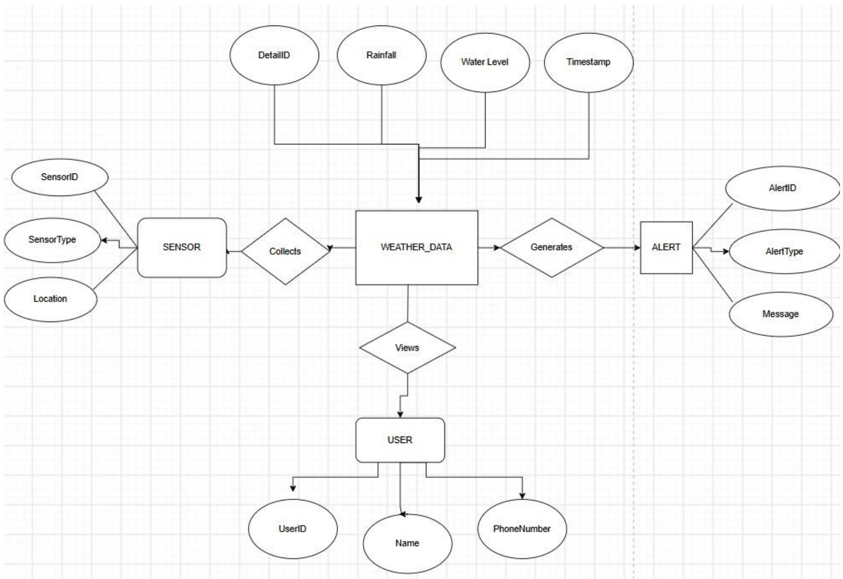
Activity Diagram



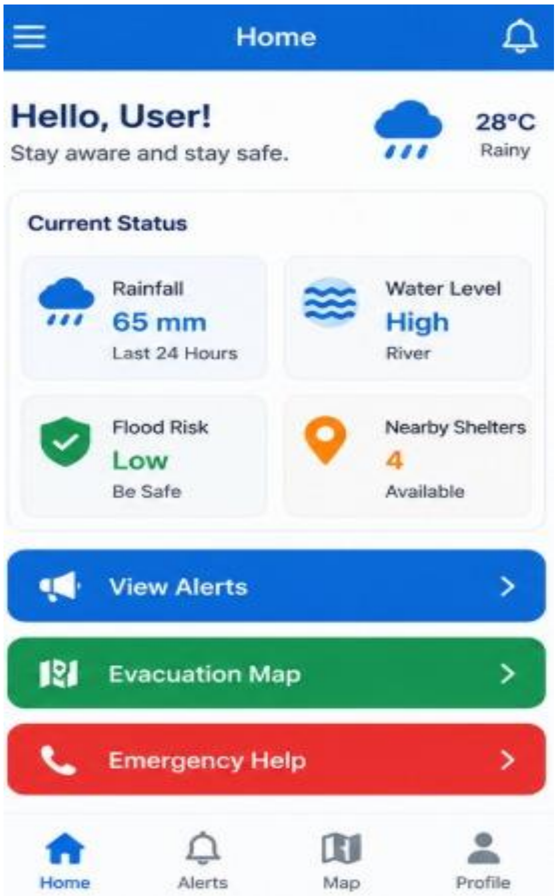
System Architecture



Database ER Diagram



UI/Wireframe:



7. Methodology / SDLC Selection

Selected Model: Agile SDLC

Why Agile?

Agile supports continuous development, rapid delivery, and frequent improvements based on user feedback. Since disaster management requirements may change depending on environmental conditions and government policies, Agile is the most suitable software development model.

Advantages

- Faster development
- Continuous testing
- Easy requirement changes
- Frequent user feedback
- Improved software quality
- Lower project risk

Development Stages

1. Requirement Gathering
2. Sprint Planning

3. Design
4. Development
5. Testing
6. Deployment
7. Maintenance

8. Implementation Plan

Phase	Duration
Requirement Gathering	1 Week
System Design	1 Week
Development	3 Weeks
Testing	1 Week
Deployment	1 Week
Maintenance	Continuous

9. Expected Outcomes

Benefits

- Early flood prediction
- Reduced property damage
- Improved emergency response
- Faster evacuation
- Better disaster management
- Increased public safety

Business Impact

- Reduced disaster management cost
- Efficient resource utilization
- Better coordination
- Improved decision making

User Benefits

- Instant flood alerts
- Safe evacuation routes
- Live flood status
- Emergency assistance

Social Impact

- Saves lives
- Reduces disaster losses

- Improves community awareness
- Supports climate resilience

SDG Contribution

Supports **SDG 13: Climate Action** by improving disaster preparedness, enhancing early warning systems, reducing climate-related risks, and protecting communities from floods.

10. Risks and Limitations

- Internet connectivity failure
- Sensor malfunction
- False flood predictions
- Cybersecurity threats
- Power outages
- Data privacy concerns
- High deployment cost
- Limited network coverage

11. Future Enhancements

- AI-based flood prediction improvements
- Drone surveillance
- Satellite image integration
- SMS and WhatsApp alerts
- Voice assistant support
- GIS-based flood mapping
- Multi-language support
- Offline emergency mode

12. Conclusion

The proposed **Flood Early Warning System** is an intelligent software solution designed to improve disaster preparedness and public safety. By integrating IoT weather sensors, machine learning, cloud computing, and mobile applications, the system provides accurate flood prediction, real-time alerts, evacuation guidance, and efficient coordination among disaster management authorities, rescue teams, and citizens. The proposed solution reduces flood-related risks and contributes significantly to **SDG 13: Climate Action**.

13. References

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